

KaarigarAI

The AI Virtual Business Manager for Marginalized Artisans

One photo. One voice note. In her own language.

Problem Statement SIH26090

AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

Ministry of Social Justice and Empowerment · Software · Miscellaneous

Team Algorhythm

THE POPULATION

64.66 lakh

handloom and handicraft artisans in India

2.72 crore

artisans enrolled under PM Vishwakarma

71%

of handloom weavers are women

9 in 10

handlooms are in rural India

The barrier is not market access. It is production.

Fairs give a burst of sales, then nothing. Between them, these artisans have no digital presence at all.

Amazon Karigar, GeM, ONDC and Flipkart Samarth all accept artisan sellers. The artisan simply cannot produce what those platforms demand: a clean photograph, a well-written description in the right language, and a competitive price.

Photography, copywriting and pricing are three separate professional skills. A single artisan is unlikely to have any of them — and low literacy means even filling in a listing form is out of reach.

So the app must substitute for a photographer, a translator, a copywriter and a pricing analyst at once — and then carry the finished listing to a buyer.

WHAT A LISTING NEEDS — AND WHY IT FAILS



A photograph

Cluttered background, poor light, wrong crop



A description

Spoken language is not the listing language



A price

No market reference, so priced flat and low

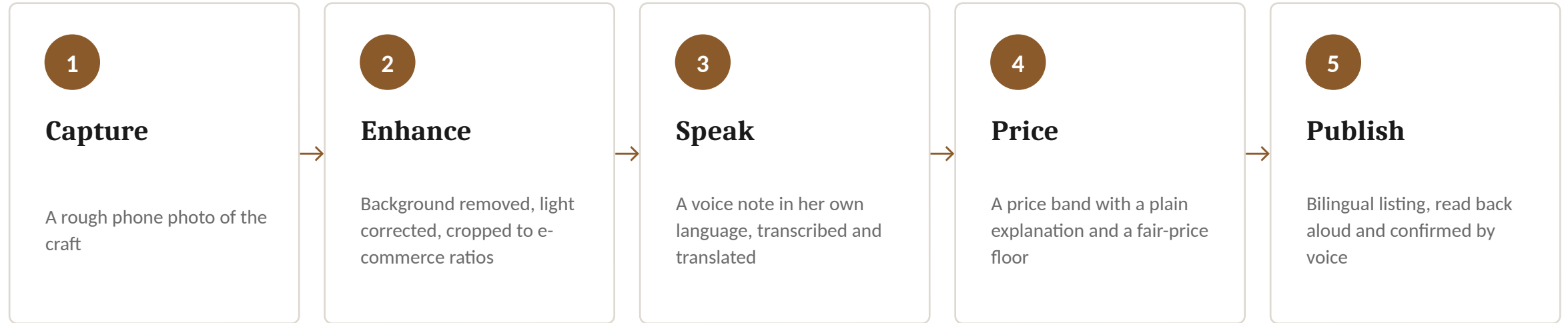


A buyer

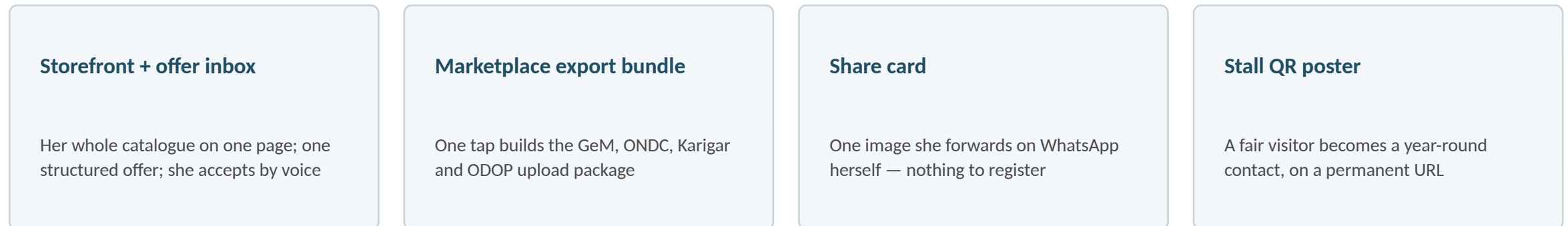
Reachable only during a fair, for three days

How it works — five steps, no typing at any point

Everything before "publish" happens on the device and survives a dead network.



THEN THE HALF MOST SOLUTIONS FORGET — GETTING IT TO A BUYER



What we build

Three features are mandated by the problem statement. Five more answer the half it asks for but rarely gets.

MANDATED — DELIVERED IN FULL, AND EXTENDED

AI Image Enhancer

rembg + OpenCV. Background removal, light and white balance, saliency crop, texture close-up for textiles.

Multilingual Auto-Cataloger

whisper.cpp + IndicTrans2 + Gemini. Craft glossary preserves authentic terms. Read back aloud before publishing.

Dynamic Pricing Assistant

Image and transcript become attributes; XGBoost returns a band; SHAP explains it; a floor prevents underpricing.

ADDED — AIMED AT MARKET LINKAGE AND LITERACY

Voice-first first run

Register without reading: tiles that speak their own name, digits only

Storefront + offer inbox

The direct answer to "connect with larger B2B buyers"

Voice order and offer status

She asks aloud and hears the answer — the loop closes

Maker story

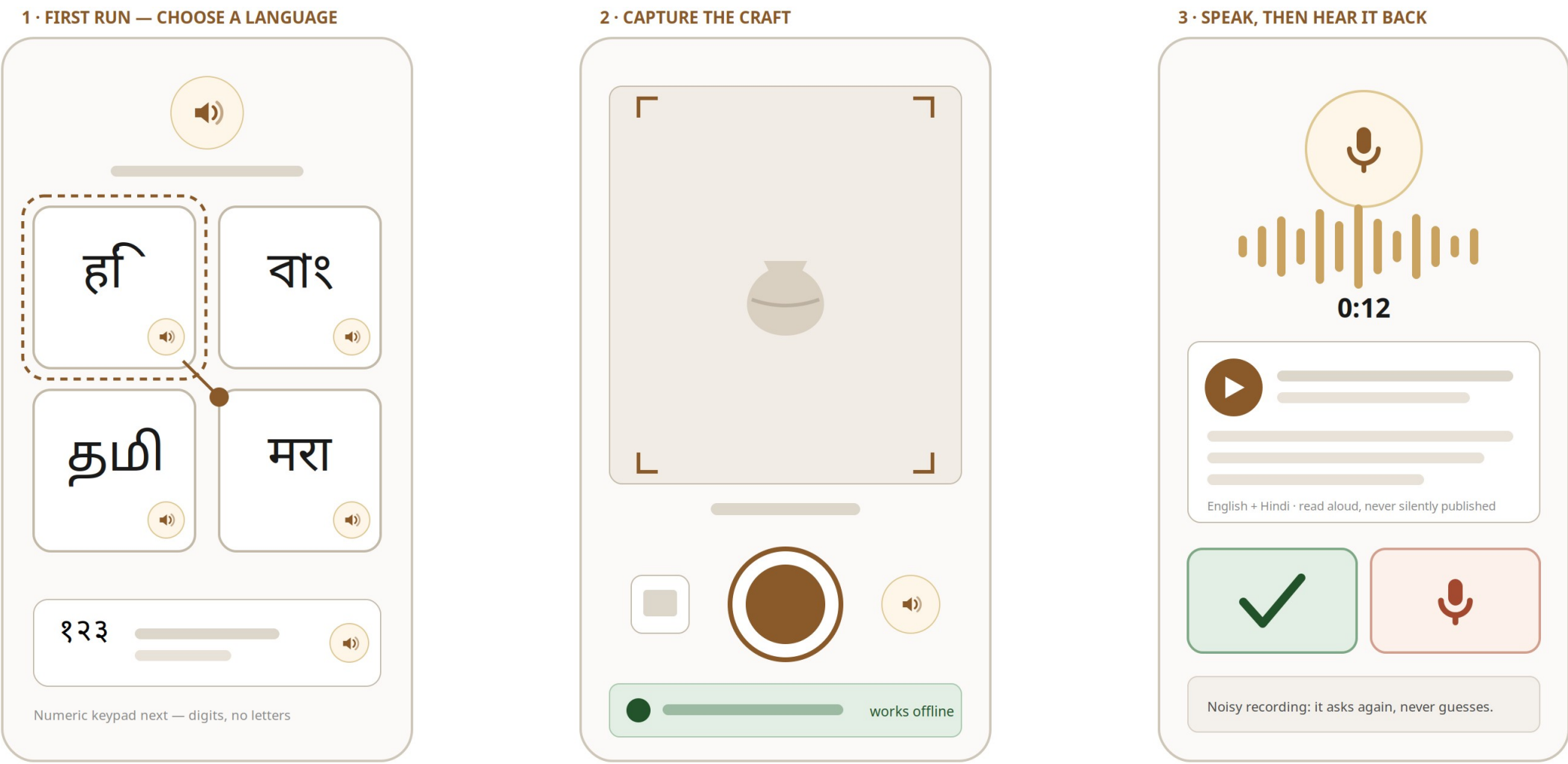
Provenance is the margin. Same voice pipeline, near-zero cost

Market linkage toolkit

Export bundle, share card, stall QR

The interface, for someone who cannot read

Grey bars mean text she is not expected to read — every one has a speaker beside it. No screen offers more than two real choices.

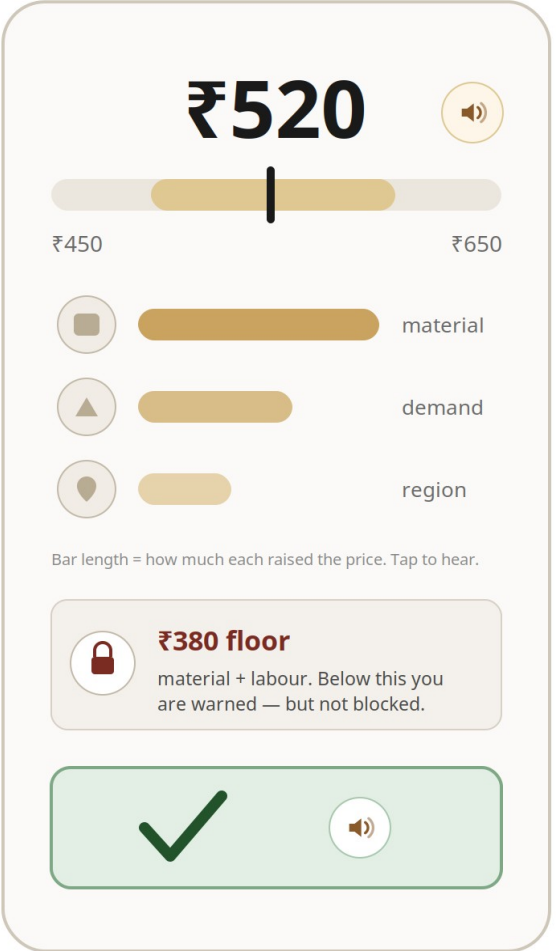


Wireframes — layout, hierarchy and touch targets. Replace with product screenshots as the build progresses.

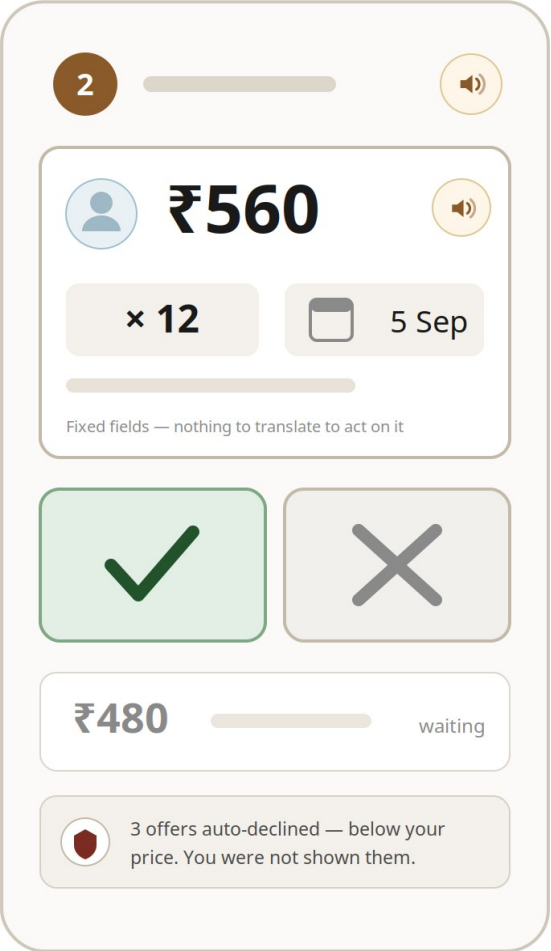
Price, offers, and what the buyer sees

SHAP as three bars rather than a table. Two buttons and one number. The buyer side is deliberately ordinary.

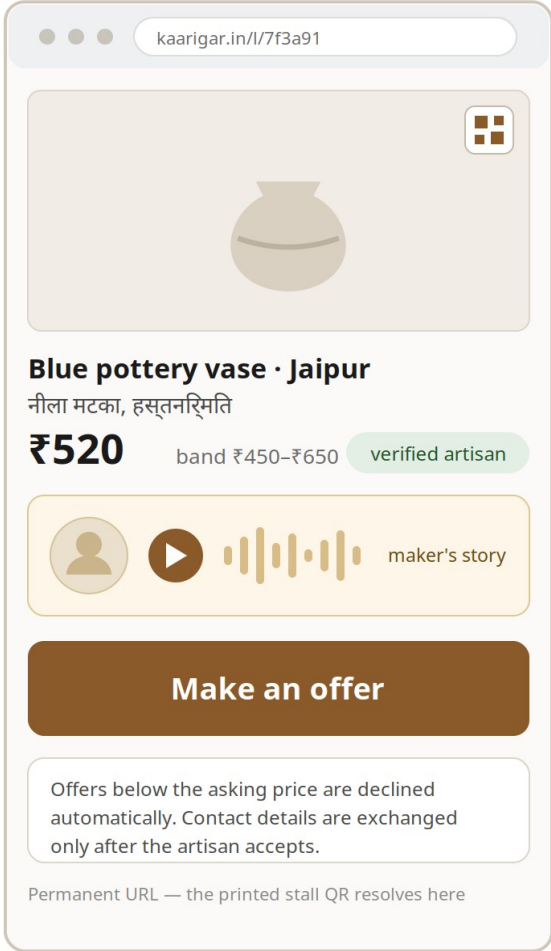
4 · WHAT IT IS WORTH, AND WHY



5 · SOMEBODY WANTS TO BUY IT



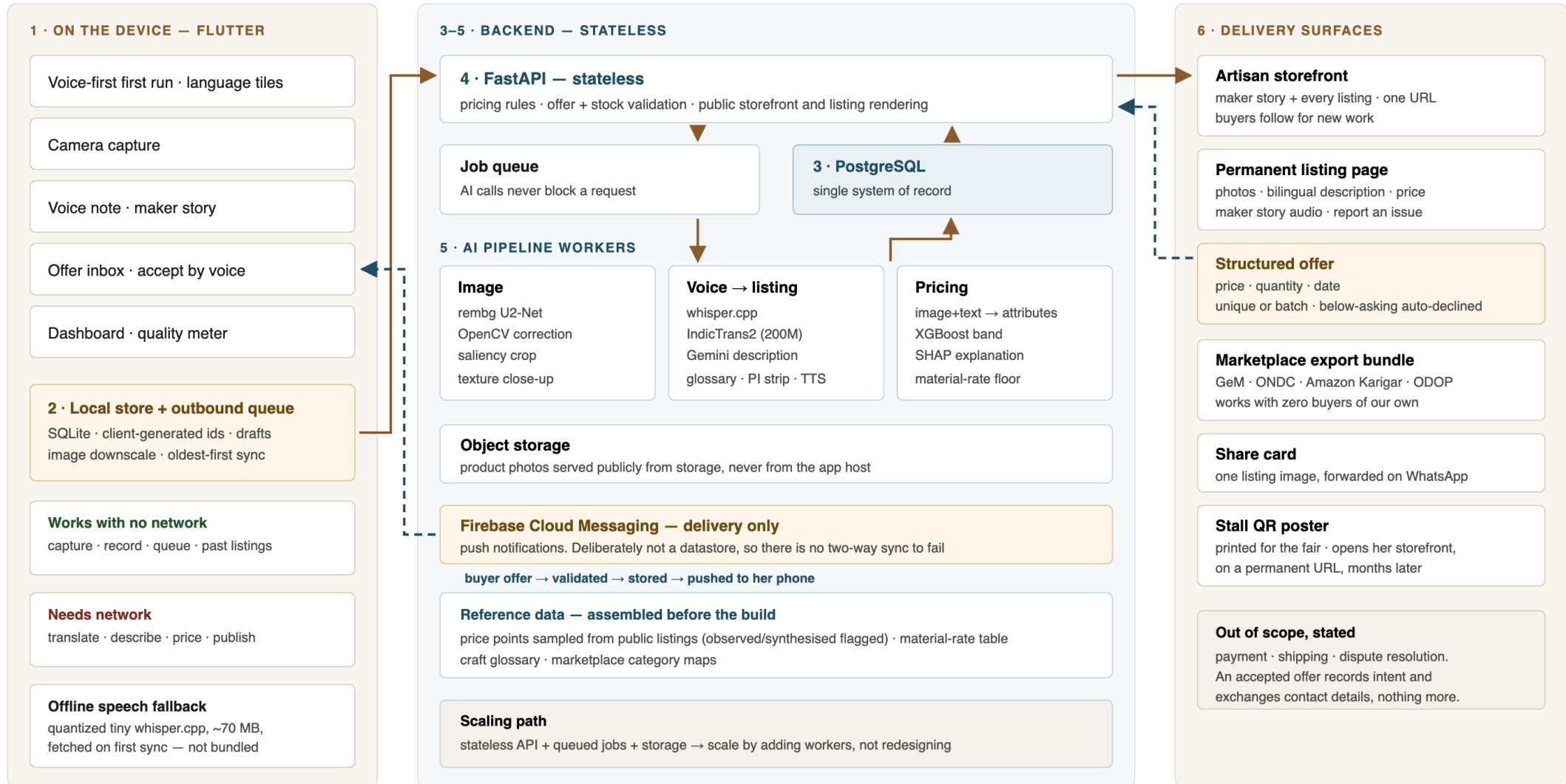
6 · WHAT THE BUYER SEES (WEB)



The only screen that assumes reading is the buyer's — boutique owners, exporters and procurement staff.

Technical approach

Six layers. One system of record. Every component free or open-source.



Feasibility and viability

Every choice is locked, and every known failure mode has a stated response.

THE STACK — NO ALTERNATIVES LISTED, BECAUSE INDECISION COSTS MORE THAN A SUBOPTIMAL CHOICE

CLIENT
Flutter

BACKEND
FastAPI, stateless

RECORD
PostgreSQL

VISION
rembg + OpenCV

SPEECH
whisper.cpp (tiny on device)

TRANSLATE
IndicTrans2 200M

TEXT
Gemini free tier

PRICING
XGBoost + SHAP

PUSH
FCM — delivery only

Risk	Response
Noisy village recording, or a mis-heard craft word	Confidence check then re-record; craft glossary fuzzy-corrects before writing
No connectivity for days	Capture, record and queue offline; oldest-first sync; client ids prevent duplicates
An unseen craft category	Wider band and stated low confidence — never a confident wrong number
Two buyers want the same piece	Stock decrements under a row lock on each acceptance; unique is the quantity-one case
A middleman operates the app for her	Account and stall QR bound to her own number; earnings visible only in her view
No venue internet or an exhausted free tier	Rehearsed offline path, local speech model, responses pre-cached

Impact and benefits

The demand side is not the constraint. The artisan making the goods is simply not in the channel.



WHAT WOULD CHANGE FOR HER

A year-round channel. Listings persist between fairs, and a printed stall QR turns a one-day visitor into a permanent contact.

A floor under her price. The app never suggests below material plus labour, and warns before she goes under it.

A reason to charge more. The maker story restores the provenance a bare product photograph destroys.

Useful on day one. The export bundle feeds marketplaces that already have buyers, so we need none of our own.

HOW WE WOULD KNOW IT WORKED

Under 2 minutes

photograph to published listing, in connected time — against hours or never today

Listings per artisan per month

and the share of months with any activity: what "year-round" means in practice

Share of listings with an offer

plus accepted price against the band midpoint — deliberately not against the floor, which would only confirm itself

The live demo — seven beats

It ends on the buyer, because market linkage is half of what the problem statement asks for.

1

First run

Register with no reading. Tap a tile that speaks its own name; hear the code read aloud.

2

Photograph

A raw, badly lit phone photo becomes a clean e-commerce image, with a texture close-up.

3

Voice note

A regional-language description becomes a bilingual listing — then it is read back for confirmation.

4

Price

A band, a plain-language explanation, and the floor visibly refusing an underpriced suggestion.

5

Offline

Disable the network. Capture continues, the listing queues, and it syncs with no duplicate.

6

The buyer

Scan the stall QR — it opens her storefront. Play the story, send an offer; she accepts by voice.

7

Linkage

One tap builds the GeM upload bundle; the stall QR prints. Close on a real scheme she may qualify for.

Payment and shipping are out of scope, and we say so: an accepted offer records intent and exchanges contact details, nothing more.

Research and references

Every figure quoted above, and every component named. All links checked.

SCHEMES AND SECTOR DATA

Handicrafts at the Heart of India's Rural Economy — PIB, 9 Dec 2025

pib.gov.in — PRID 2200859

PM Vishwakarma enrolment, Rajya Sabha reply, 1 Dec 2025

newsonair.gov.in

Fourth All India Handloom Census 2019-20, Ministry of Textiles

archive.org / handlooms.nic.in

PM Vishwakarma scheme portal

pmvishwakarma.gov.in

Office of the DC (Handicrafts) and DC (Handlooms)

handicrafts.nic.in · handlooms.nic.in

MARKETPLACES AND POLICY

Government e-Marketplace · ONDC · Amazon Karigar

gem.gov.in · ondc.org · amazon.in/karigar

One District One Product, DPIIT — PIB, 23 Jan 2026

odop.dpiit.gov.in

DPDP Act 2023 and DPDP Rules, notified 14 Nov 2025

meity.gov.in · indiacode.nic.in

OPEN-SOURCE COMPONENTS

AI4Bharat IndicTrans2 — models and paper

github.com/AI4Bharat/IndicTrans2 · [arXiv 2305.16307](https://arxiv.org/abs/2305.16307)

whisper.cpp — local inference; Whisper paper

github.com/ggml-org/whisper.cpp · [arXiv 2212.04356](https://arxiv.org/abs/2212.04356)

rembg background removal; U²-Net paper

github.com/danielgatis/rembg · [arXiv 2005.09007](https://arxiv.org/abs/2005.09007)

XGBoost; SHAP paper

xgboost.readthedocs.io · [arXiv 1705.07874](https://arxiv.org/abs/1705.07874)

OpenCV · Flutter · FastAPI · PostgreSQL

opencv.org · flutter.dev · fastapi.tiangolo.com

Firebase Cloud Messaging · Google AI Studio

firebase.google.com · aistudio.google.com

A NOTE ON SOURCES

Where a statistic exists in both a primary government release and a secondary summary, the primary is cited. Two earnings figures come from a secondary summary of ministry material and are presented as evidence about the sector, not as predictions for this application.

We built the app to be the professional photographer, translator, copywriter and pricing analyst an artisan could never afford to hire — and then to find her a buyer.

One photo. One voice note.